

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

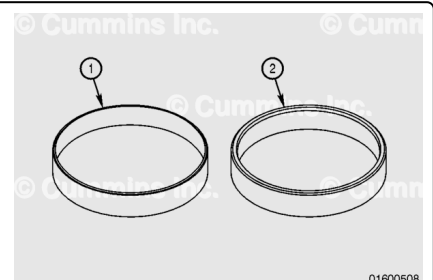
- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/56/56-013-009.html>)
- Remove the front gear housing. Refer to Procedure 001-033 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-033-tr.html>)



Remove

⚠ CAUTION ⚠

Do not allow the chisel to cut through the sleeve and damage the gear cover.



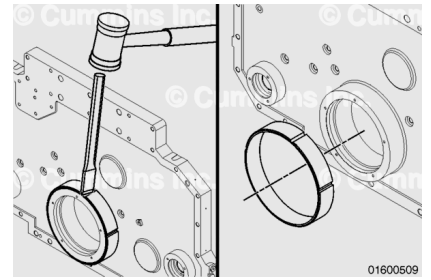
Note : Determine the sleeve type prior to removal.

There are two gear cover trunion sleeve types:

1. thin 3 mm [0.12 in]
2. thick 12 mm [0.47 in]

⚠ WARNING ⚠

Wear appropriate eye and face protection when using a chisel and mallet. Flying debris and dirt can cause personal injury.



Note : For the thin sleeve, the objective is to deform the sleeve, not to cut it into 2 pieces.

Steps for the thin gear cover trunion sleeve:

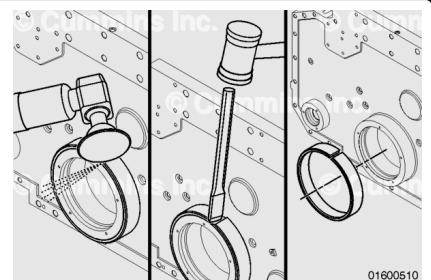
Use a chisel and mallet. Cut approximately 3/4 of the thickness of the sleeve at 2 or 3 points.

The sleeve will lose its shape and the press fit will be reduced.

Remove and discard the sleeve.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using a grinder, chisel, and mallet. Flying debris and dirt can cause personal injury.



Steps for the thick gear cover trunion sleeve:

Use a suitable grinding tool to reduce an area of the sleeve to approximately 3 mm [0.12 in] thickness. Take care **not** to damage the gear cover.

Use a chisel and mallet to cut into the ground area.

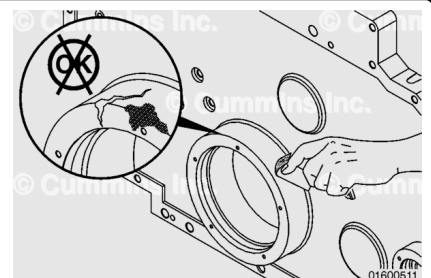
The sleeve will split and the press fit will be reduced.

Remove and discard the sleeve.

Inspect for Reuse

Check the gear cover for cracks or other damage. If the gear cover is damaged, it **must** be replaced.

Use emery paper. Remove any nicks or burrs from the outside diameter of the trunion.

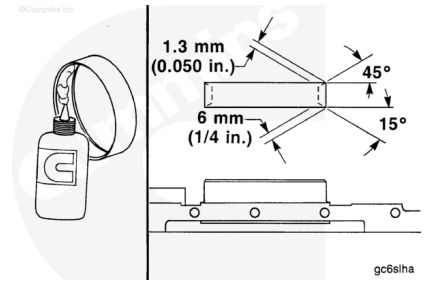


Install

Use Loctite® 609, or equivalent. Apply a smooth coating to the inside diameter of the sleeve.

The side of the sleeve with the widest 6.1 mm [0.24 inch] chamfer **must** be positioned toward the cover. Position the sleeve on the cover.

Note : The thick sleeve has an identical chamfer on both sides and can be fitted in either orientation.

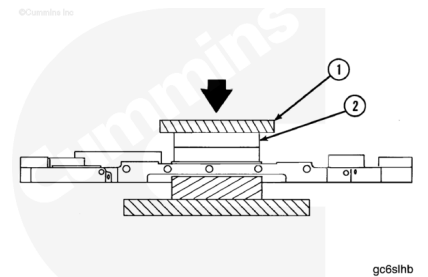


Use a hydraulic press. Support the gear cover as illustrated.

Use a flat steel plate (1). Push the sleeve (2) until the plate (1) touches the gear cover.

Check the sleeve for cracks or torn areas.

If the sleeve cracks or tears during installation, the sleeve **must** be replaced. Make sure there are no burrs on the cover. Make sure the sleeve is positioned straight with the cover before installing it with the plate and press.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the front gear housing Refer to Procedure 001-033 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-033-tr.html)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/56/56-013-009.html)
- Operate the engine and check for proper operation.

